

SE LAB 1

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Deliverable 1:

Requirements Table

Multi-Vendor Artisan E-Commerce Marketplace (Problem Statement #31)

Actors: Shopper, Artisan Vendor, Payment Gateway

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall split customer cart payments at checkout, allocating respective item earnings to multiple independent vendor accounts after deducting a 5% platform fee. (Given)	High	Pass: Split payout calculations balance to total cart value. Fail: Payout calculation discrepancies.	Correct, transparent payment splitting is the platform's core financial function and a trust/legal requirement for multi-party marketplaces.
FR-002	Functional	The system shall allow an Artisan Vendor to register and configure a storefront (shop name, banner, bio, policies) before listing any products.	High	Pass: A vendor with an unconfigured storefront cannot publish a product listing. Fail: Products appear live under an incomplete/unregistered storefront.	A storefront is the vendor's identity on the platform; buyers need it to trust a seller, and payouts must route to a registered account.
FR-003	Functional	The system shall allow an Artisan Vendor to create, update, and deactivate product listings, including price, stock quantity, images, and category.	High	Pass: Stock quantity decrements automatically on order confirmation and the listing is auto-hidden at zero stock. Fail: Orders can be placed against out-of-stock or deactivated listings.	Accurate catalog state prevents overselling handmade/limited-quantity goods, a core artisan-marketplace risk.
FR-004	Functional	The system shall let a Shopper add items from multiple different vendors into a single	High	Pass: A single checkout can contain items from 2+ vendors and produces one combined	Multi-vendor carts are the core value proposition distinguishing a

		cart and complete checkout in one transaction.		order confirmation. Fail: Shopper is forced into separate checkouts per vendor.	marketplace from a single-seller store.
FR-005	Functional	The system shall allow a Shopper to track order status (placed, shipped, delivered) per vendor within a single order, and notify them of status changes.	Medium	Pass: Shopper sees independent status per vendor sub-order and receives a notification within 1 minute of a status change. Fail: Status updates are missing, delayed, or merged incorrectly across vendors.	One order can span multiple vendors with independent fulfillment timelines, so per-vendor tracking avoids shopper confusion and support tickets.

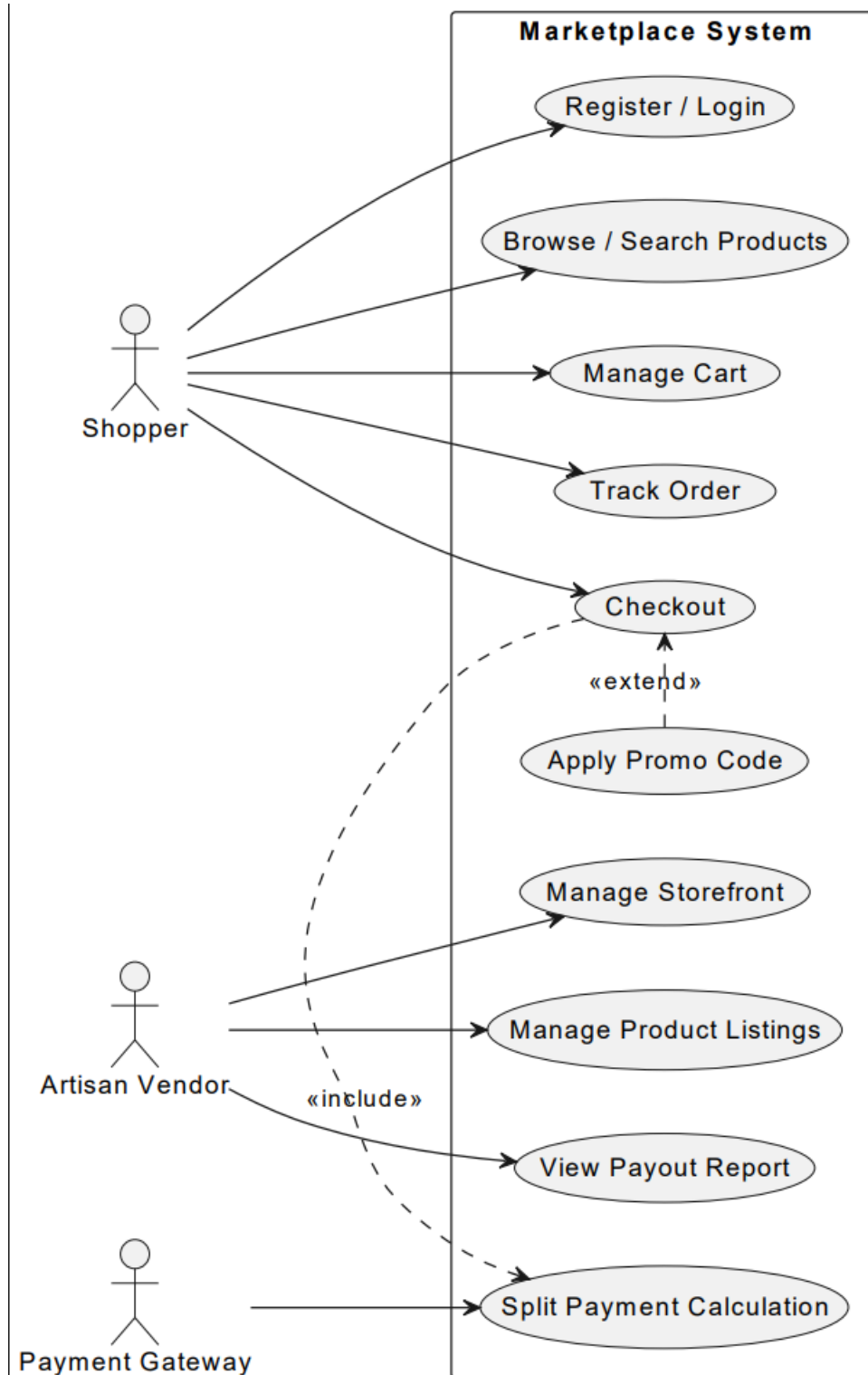
Nonfunctional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	The product catalog must support high-resolution image rendering with CDN caching delivering load times < 500 ms. (Given)	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Load times exceed target under peak load.	Product photos drive purchase decisions for artisan goods, so image latency directly affects browsing experience and conversion.
NFR-002	Nonfunctional	All payment processing and split-payout data must be handled through a PCI-DSS compliant payment gateway, with no raw card data stored on platform servers.	High	Pass: Security audit confirms no card data at rest on platform infrastructure and the gateway integration is PCI-DSS certified. Fail: Any card data found stored or transmitted outside the certified gateway.	Handling funds for multiple third-party vendors raises the platform's regulatory and liability exposure, making payment compliance non-negotiable.

Deliverable 2:

Use-Case Diagram

Multi-Vendor Artisan E-Commerce Marketplace



Deliverable 3:

Use-Case Flow Specification

Use Case: *Checkout (with Split Payment)*

Actor(s): Shopper (primary), Artisan Vendor (secondary — receives payout), Payment Gateway (secondary — authorizes and processes payment)

Related use cases: «include» Split Payment Calculation · «extend» Apply Promo Code

Preconditions

1. The Shopper is logged in.
2. The cart contains at least one item, sourced from one or more Artisan Vendors.
3. All items in the cart are currently in stock.
4. The Shopper has a valid, verified payment method on file or provided at checkout.

Postconditions

Success: An order record is created with one sub-order per vendor; the Shopper's payment method is charged the total cart value; funds are split and credited to each vendor's payout account (minus 5% platform commission) via the Payment Gateway; stock quantities are decremented; the Shopper and each involved Vendor receive an order confirmation notification.

Failure: No charge is made, no order record is created, and the cart remains unchanged.

Main Success Scenario

1. The Shopper selects “Proceed to Checkout” from the cart.
2. The system displays an order summary grouped by vendor, including item subtotals, shipping, and grand total.
3. The Shopper confirms or selects a payment method.
4. The Shopper submits the order.
5. The system validates that all items are still in stock and prices are unchanged.
6. The system invokes Split Payment Calculation («include»), which computes each vendor's share of the total after deducting the 5% platform commission.
7. The system sends the charge request to the Payment Gateway for the full cart total in a single transaction.
8. The Payment Gateway authorizes the charge and confirms back to the system.
9. The system credits each vendor's payout account with their computed share and creates the order record with one sub-order per vendor, decrementing stock for each purchased item.
10. The system displays an order confirmation to the Shopper and sends notifications to the Shopper and each involved Vendor.

Alternate Flow: Item Goes Out of Stock Between Cart and Checkout

Trigger: At step 5, the system detects that one or more cart items are no longer in stock.

1. The system halts the transaction before any charge is made.
2. The system flags the affected item(s) in the order summary and displays an out-of-stock message.
3. The system removes the affected item(s) from the cart, or offers the Shopper an option to wait for restock if the vendor allows backorders.
4. The Shopper reviews the updated cart and either resubmits checkout with the remaining items (returning to step 4 of the main flow) or exits checkout.

Postcondition (alternate): No charge is made; the cart reflects only in-stock items; no order record is created for the affected item(s).